From Cyclotrons to Synchrotrons

How Circular Accelerators Work—and Why We Need Both

Supplementary to Mann, Section 7.3

1 The Central Idea: Reuse One Voltage Source

A linear accelerator (linac) uses a *sequence* of accelerating gaps, each contributing energy $\Delta E = qV$ to the beam. To reach high energies you need many gaps, which means a very long (and expensive) machine.

The circular trick

What if we bent the beam into a circle so it passed through the *same* accelerating gap over and over? Each revolution adds another qV of energy, so after N turns:

$$K = NqV$$

One modest voltage source, applied N times, replaces N separate accelerating structures. This is the idea behind every circular accelerator.

The challenge is *bending* the beam. A charged particle in a uniform magnetic field $\vec{B}$ (perpendicular to its velocity) moves in a circle of radius

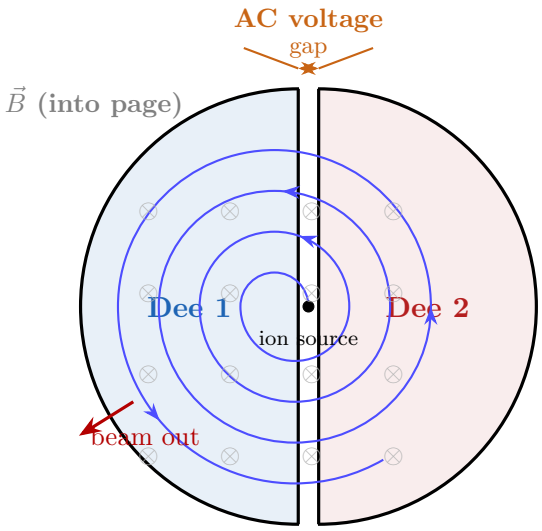
$$r = \frac{p}{qB} = \frac{\gamma m v}{qB} \tag{1}$$

where $p = \gamma m v$ is the relativistic momentum. This is the **cyclotron radius** (or Larmor radius). The key insight: as the particle gains energy, p increases, so r increases—the orbit spirals outward.

2 The Cyclotron (Lawrence, 1930)

2.1 Physical Layout

A cyclotron consists of two hollow, D-shaped metal electrodes—called **dees**—placed face-to-face inside a large electromagnet.



2.2 How It Works, Step by Step

1. **Ion source at center.** A source at the center of the machine produces ions (e.g. protons) essentially at rest.

2. **Magnetic field bends the path.** A large electromagnet produces a uniform field $\vec{B}$ perpendicular to the plane of the dees (into the page in the figure). This bends the ion into a semicircle inside one dee.

3. **Dees are field-free regions.** The dees are hollow metal conductors. By Gauss's law, there is *no electric field inside* a conductor, so the ion coasts at constant speed along its semicircular arc. The only role of $\vec{B}$ here is to curve the trajectory.

4. **The gap is the accelerating region.** An alternating (AC) voltage is applied between the two dees. Every time the ion crosses the gap, it sees an electric field pointing in the direction of its motion and gains energy $\Delta K = qV$.

5. **The orbit spirals outward.** After each acceleration, the ion has more momentum, so from Eq. (1), its orbital radius $r = p/(qB)$ is larger. The trajectory is a spiral of increasing radius.

6. **Frequency matching (the key constraint).** The AC voltage must flip polarity at exactly the right moment—when the ion arrives at the gap. This works because, non-relativistically, the **cyclotron frequency** is independent of speed:

Cyclotron Frequency (Non-Relativistic)

$$f_{\text{cyc}} = \frac{qB}{2\pi m} \qquad \text{(independent of } v \text{ or } r\text{)}$$

This is why a *fixed-frequency* AC source works: no matter how fast the ion is going or how large its orbit, it always takes the same time to complete a semicircle.

Why is the frequency constant?

As the ion speeds up, its orbit gets bigger. It travels a longer path, but it's moving faster—and these two effects *exactly cancel* (non-relativistically). The circumference scales as $r \propto v$, so the period $T = 2\pi r/v$ is constant. This beautiful cancellation is what makes the cyclotron practical.

2.3 Derivation of the Cyclotron Frequency

The magnetic force provides the centripetal acceleration:

$$qvB = \frac{mv^2}{r} \quad \Longrightarrow \quad r = \frac{mv}{qB}$$

The period of one full orbit is:

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

So the frequency is:

$$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

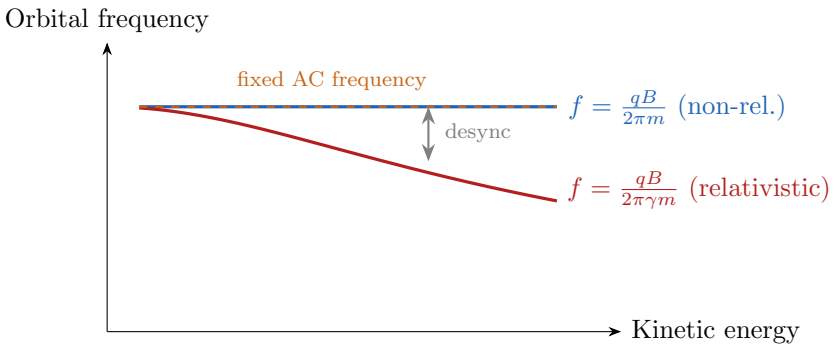
and the angular frequency is $\omega_{\text{cyc}} = qB/m$. Note: *no dependence on v or r*—this is the whole reason the cyclotron works with a fixed-frequency oscillator.

2.4 The Relativistic Problem: Why Cyclotrons Top Out at ∼20 MeV

Relativistically, $r = \gamma mv/(qB)$, and the period becomes:

$$T = \frac{2\pi\gamma m}{qB}$$

As γ increases, the ion takes *longer* to complete each orbit—it falls behind the fixed AC frequency. The ion arrives at the gap out of phase and starts *decelerating* instead of accelerating.



For protons, relativistic effects become important around $K \sim 20\,\text{MeV}$ ($\gamma \approx 1.02$). This is the practical ceiling for a fixed-frequency cyclotron.

Partial fix: the synchrocyclotron

One can *slowly decrease* the AC frequency as the beam spirals out, tracking the dropping cyclotron frequency. This is called a **synchrocyclotron**. It works, but it can only accelerate one "pulse" of ions at a time (since the frequency is tuned to one particular orbit radius at any instant), which greatly reduces beam intensity.

3 The Synchrotron: The Modern Solution

3.1 Key Idea

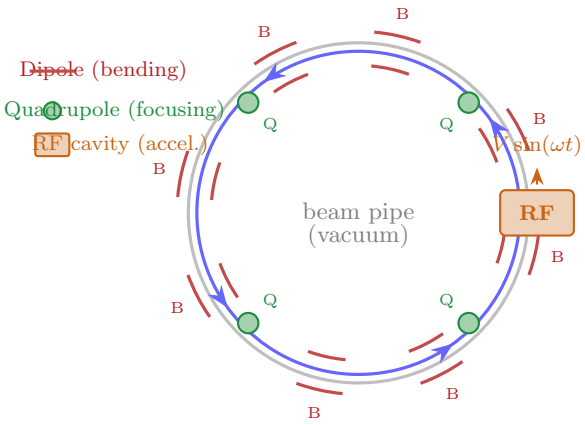
Rather than letting the orbit spiral outward, the synchrotron keeps the beam at a **fixed radius** by *increasing the magnetic field as the particle accelerates*:

Synchrotron Condition

$$r = \frac{p}{qB(t)} = \text{constant} \quad \Longrightarrow \quad B(t) \propto p(t)$$

As the momentum p rises, B is ramped up in lockstep to keep r constant.

This has an enormous practical advantage: magnets are only needed *along the ring*, not filling the entire interior.



3.2 Components of a Synchrotron

1. Dipole bending magnets steer the beam around the ring. They produce a field $B(t)$ that ramps with the beam momentum to maintain constant radius.

2. RF (radio-frequency) cavities provide the accelerating voltage. The oscillating electric field is timed so the beam bunch arrives at the cavity at the right phase to be accelerated on every pass. The RF frequency must increase during acceleration to match the increasing revolution frequency:

$$f_{\text{RF}}(t) = \frac{v(t)}{2\pi r}$$

As $v \to c$, the frequency asymptotes to $f \approx c/(2\pi r)$ and barely changes at ultrarelativistic energies—a major simplification.

3. Quadrupole magnets focus the beam transversely (this is the FODO lattice we discussed earlier).

4. Vacuum beam pipe. The beam circulates in an evacuated tube ($\sim 10^{-9}$ torr) to avoid scattering off residual gas.

3.3 Why the Synchrotron Solves the Relativistic Problem

In a cyclotron, the magnetic field is fixed and the frequency is fixed—relativity breaks the synchronization. In a synchrotron, *both* B and f_{RF} are dynamically adjusted:

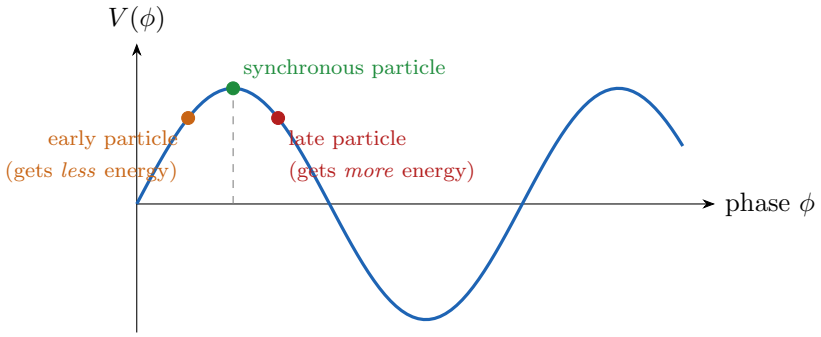
	Cyclotron	**Synchrocyclotron**	**Synchrotron**
B field	fixed	fixed	ramped with p
$f_{\mathrm{AC/RF}}$	fixed	slowly decreased	ramped with v
Orbit radius	spirals out	spirals out	**fixed**
Magnet coverage	entire area	entire area	**ring only**
Max energy (protons)	$\sim$20 MeV	$\sim$700 MeV	TeV scale

The cost advantage

A cyclotron needs a solid magnet filling the entire area πR^2 of the orbit. A synchrotron only needs magnets along the circumference $2\pi R$. For a ring of radius 4 km (like the LHC), filling the interior with magnet iron would be absurd. The synchrotron design is what makes machines like the LHC, Tevatron, and RHIC feasible.

3.4 Phase Stability

A subtle but essential feature: the synchrotron is *self-correcting* against small energy errors. This is called **phase stability** (discovered independently by Veksler and McMillan, 1944–45).



The RF voltage oscillates sinusoidally. The ideal ("synchronous") particle arrives at the peak and gets exactly the right energy boost. A particle that arrives *late* sees a *higher* voltage (it's on the rising edge), so it gets *more* energy, moves to a slightly larger orbit, and catches up. An *early* particle gets less energy and falls back. The result is that particles oscillate stably around the synchronous phase—they are trapped in "RF buckets." This is why synchrotron beams come in discrete **bunches** rather than a continuous stream.

4 Cyclotron vs. Synchrotron: When to Use Which

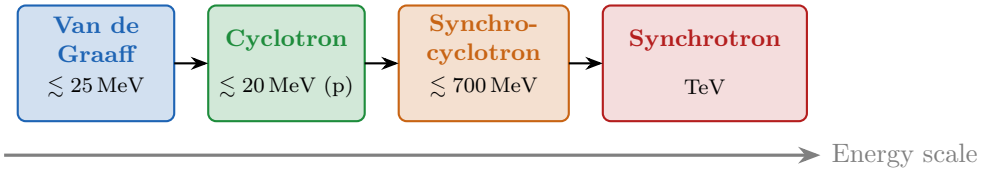
Cyclotrons have not disappeared. They are compact, produce *continuous* (not bunched) beams, and are ideal for:

- Medical isotope production (PET tracers, ^{18}F, ^{11}C)
- Proton therapy for cancer treatment ($\sim$250 MeV)
- Low-energy nuclear physics

Synchrotrons are needed whenever energy must go beyond a few hundred MeV:

- High-energy particle physics (LHC: 6.5 TeV per beam)
- Synchrotron light sources (X-ray beamlines)
- Heavy-ion colliders (RHIC)

5 Putting It All Together



Summary of Key Equations

$$\text{Cyclotron radius:} \quad r = \frac{\gamma m v}{qB} \tag{2}$$

$$\text{Cyclotron frequency (non-rel.):} \quad f = \frac{qB}{2\pi m} \tag{3}$$

$$\text{Synchrotron condition:} \quad B(t) = \frac{p(t)}{qr} \quad (r = \text{const.}) \tag{4}$$

$$\text{Energy per pass:} \quad \Delta E = qV_{\text{RF}} \sin\phi_s \tag{5}$$

where ϕ_s is the synchronous phase.